Vahab Jabrayilov

vjabrayilov@cs.columbia.edu
vjabrayilov.github.io | github.com/vjabrayilov

Systems engineer focused on ML performance: accelerators (TPU, GPU), low-latency networking, and scheduling.

Education

2023 - 2027 (Expected) **Ph.D., Computer Science**, *Columbia University*, New York, NY, USA
Advisor: Prof. Kostis Kaffes

2023 - 2025 **M.Sc., Computer Science**, *Columbia University*, New York, NY, USA

2019 - 2023 **B.Sc., Computer Engineering**, *Middle East Technical University*, Ankara, Turkey
GPA: 3.8/4.0 Ranking: 7/305

Work Experience

May 2026 - Present **Google**, *Student Researcher*, New York, NY and Sunnyvale, CA, USA
- Optimize TPU fleet utilization with runtime and compiler performance work (TPU Runtime team).

Aug 2023 - Present **Columbia University**, *Graduate Research Assistant*, New York, NY, USA
- Built Fence, a runtime that adds fault isolation to dynamic spatial sharing on unmodified NVIDIA GPUs for LLM serving.
- Built Machnet, a C++ DPDK messaging library for public cloud VMs that handles 750K RPS on Azure at 61us tail latency.
- Built Rorke, a microsecond-scale VM scheduler in Rust and eBPF.

Jun-Sep 2024 **Microsoft Research**, *Research Software Engineer Intern*, Redmond, WA, USA
- Built a Rust DPDK packet processor that handles millions of flow logs/sec with deterministic sub-ms latency for Azure's real-time data ingestion pipeline (Networking Research Group).

Jun-Aug 2023 **EPFL**, *Research Intern*, Lausanne, Vaud, Switzerland
- Analyzed metastable failures in distributed systems and designed mitigations.

Selected Publications

ASPLOS, 2026 Enabling Fast Networking in the Public Cloud
Alireza Sanaee*, **Vahab Jabrayilov***, Ilias Marinos, Farbod Shahinfar, Divyanshu Saxena, Gianni Antichi, Kostis Kaffes (* equal contribution)

Under Review Fence: Safe Dynamic Spatial Sharing for GPUs
Vahab Jabrayilov, Pulak Mehrotra, Chenxi Huang, Jason Nieh, Junfeng Yang, Kostis Kaffes

Under Review Rorke: Black-box Virtual Machine Scheduling for the Public Cloud
Vahab Jabrayilov, Teng Jian, Jason Nieh, Kostis Kaffes

NeurIPS MLForSys Workshop, 2025 An Expert in Residence: LLM Agents for Always-On Operating System Tuning
Georgios Liargkovas, **Vahab Jabrayilov**, Hubertus Franke, Kostis Kaffes

VLDB, 2025 HoliPaxos: Towards More Predictable Performance in State Machine Replication
Zhiying Liang, **Vahab Jabrayilov**, Abutalib Aghayev, Aleksey Charapko

ACM SoCC, 2024 Rethinking the Networking Stack for Serverless Environments: A Sidecar Approach
Vishwanath Seshagiri, Abhinav Gupta, **Vahab Jabrayilov**, Avani Wildani, Kostis Kaffes

ICDCS, 2025 Practical Considerations for Implementing State Machine Replication in the Cloud
Zhiying Liang, **Vahab Jabrayilov**, Abutalib Aghayev, Aleksey Charapko

Skills

Languages C/C++, Rust, Python, Assembly (x86, Arm)
ML & Accelerators CUDA, PyTorch, JAX/XLA, vLLM, NCCL, MLIR, LLVM, Nsight, NVIDIA MIG/MPS
Systems Linux kernel, KVM, eBPF, DPDK, RDMA, InfiniBand, perf, bpftrace

Honors

2018 ACM ICPC semi-final Honorable Mention

2017 Rank 1 of ∼70,000, Azerbaijan National University Entrance Exam